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A CUSUM control chart for monitoring the variance when parameters are estimated

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ABSTRACT

CUSUM control chart has been widely used for monitoring the process variance. It is usually used assuming that the nominal process variance is known. However, several researchers have shown that the ability of control charts to signal when a process is out of control is seriously affected unless process parameters are estimated from a large in-control Phase I data set. In this paper we derive the run length properties of a CUSUM chart for monitoring dispersion with estimated process variance and we evaluate the performance of this chart by comparing it with the same chart but with assumed known process parameters.

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1. Introduction

The cumulative sum (CUSUM) control chart has been widely used to monitor the variability of processes. It was introduced by Page (1954), in the case of the mean, and it is used to detect persistent shifts in a process. The use of the CUSUM chart for monitoring the variance has been studied for the first time by Page (1963). Tuprah and Ncube (1987) compared the performance of CUSUM charts with sample ranges and sample standard deviations. Chang and Gan (1995) proposed the use of the logarithmic transformation of the sample variance $\log(S^2)$ in a CUSUM chart for monitoring the process dispersion. Srivastava (1997) presented a method based on the logarithm of the likelihood ratio of the probability density functions under two competing hypotheses.

The effect of estimating the parameters in-control charts is a problem that has drawn the attention of many researchers for the last years (see e.g. Chen, 1997, 1998; Albers and Kallenberg, 2004; Albers et al., 2004). It has been shown for a number of control charts that when the process parameters are estimated, the performance of the control chart differs significantly from the known parameter case. This fact is attributed to the extra variability of the estimators that is imputed in the control chart. In the case of the CUSUM chart for the mean Jones et al. (2004) evaluated its performance when the parameters are estimated. More recently, in a review paper, Jensen et al. (2006) mention that the case of estimating the parameters in a CUSUM chart for the variance has not been studied, a fact that motivated this work.

In Section 2, we present the CUSUM chart for monitoring the variance and we introduce two one-sided (downward and upward) modified versions used in the remaining of the paper. In Section 3, we compute the exact run length distribution of the proposed schemes along with the two first moments using two different approaches: integral equations and Markov

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chain. The effect of estimating the variance on the performance of the proposed one-sided schemes is presented in Section 4. An example of use of the proposed CUSUM chart for the variance is presented in Section 5. Finally, we give some conclusions.

2. CUSUM control charts for monitoring the variance

Assume that our aim is to monitor the variance of a process using a CUSUM chart. An indispensable assumption for the development of a (Phase II) CUSUM control chart is that the process variance is known. In practice, the unknown in-control process variance σ_0^2 has to be estimated from a (Phase I) data set composed of m samples of size n $X_{i,j}$, $i=1,\ldots,m$, $j=1,\ldots,n$. If we assume that $X_{i,j}\sim N(\mu_0,\sigma_0)$, where μ_0 is the in-control process mean, σ_0^2 is the in-control process variance, and the samples are simple random samples themselves and are independent of each other, then an unbiased estimate $\hat{\sigma}_0^2$ of σ_0^2 is

$$\hat{\sigma}_0^2=\frac{1}{m(n-1)}\sum_{i=1}^{m}\sum_{j=1}^{n}(X_{i,j}-\overline{X}_i)^2,$$

where $\overline{X}_i$ is the sample mean corresponding to the ith sample, i.e.

$$\overline{X}_i=\frac{1}{n}\sum_{j=1}^{n}X_{i,j}.$$

Let us denote by $Y_{i,1},\ldots,Y_{i,n}$ the Phase II samples obtained at times $i=1,2,\ldots$. Assume also that $Y_{i,j}\sim N(\mu_0,\tau\sigma_0)$ where $\tau>0$ is a constant reflecting the shift in dispersion. The case $\tau=1$ corresponds to the in-control state for the process variability; the opposite case ($\tau\neq1$) pinpoints a shift in the process variability. A useful note is that when $\tau<1$ the out-of-control process condition corresponds to a reduction in process variability, which is usually the outcome of a corrective action on the process itself. On the other hand, when $\tau>1$ the out-of-control process condition may be the result of a deterioration in the process performance which increases the process variability. Let $\overline{Y}_i$ and S_i^2 be the sample mean and sample variance of $Y_{i,1},\ldots,Y_{i,n}$, i.e.

$$S_i^2=\frac{1}{n-1}\sum_{j=1}^{n}(Y_{i,j}-\overline{Y}_i)^2 \quad \text{with } \overline{Y}_i=\frac{1}{n}\sum_{j=1}^{n}Y_{i,j}.$$

Chang and Gan (1995) introduced an upward CUSUM control chart for detecting increases in the process variance using the scheme

$$U_0=u_0,$$

$$U_i=\max(U_{i-1}+\ln(S_i^2)-k_u,0),\quad i=1,2,\ldots, \tag{1}$$

where $0\le u_0<UCL$ and k_u is a constant. This chart signals when U_i crosses the upper control limit (UCL) which is a constant that is when $U_i\ge UCL$. The corresponding downward CUSUM control chart for detecting decreases in the process variance is given by the scheme

$$D_0=d_0,$$

$$D_i=\min(D_{i-1}+\ln(S_i^2)+k_d,0),\quad i=1,2,\ldots, \tag{2}$$

where $LCL<d_0\le0$ and k_d is a constant. Using the same way of thinking the downward CUSUM chart signals when $D_i\le LCL$ where LCL is the lower control limit which is also a constant. The charts defined in Eqs. (1) and (2) can be used only with subgrouped data (i.e. $n>1$) therefore in this paper we confine ourselves to this case.

In this paper we propose modified versions of the schemes in Eqs. (1) and (2). Specifically, the proposed upward CUSUM chart is

$$G_0^+=g_0^+,$$

$$G_i^+=\max(G_{i-1}^+ +\ln(S_i^2/\hat{\sigma}_0^2)-k,0),\quad i=1,2,\ldots \tag{3}$$

and the corresponding downward CUSUM chart is

$$G_0^-=g_0^-,$$

$$G_i^-=\min(G_{i-1}^- +\ln(S_i^2/\hat{\sigma}_0^2)+k,0),\quad i=1,2,\ldots, \tag{4}$$

where $0\le g_0^+<h$ ($-h<g_0^-\le0$) and k is a constant (this constant in the CUSUM literature is usually called reference value). The upward (downward) CUSUM chart signals at the first i that $G_i^+>h$ ($G_i^-<-h$). If $g_0^+>0$ ($g_0^-<0$) then the upward (downward) chart has a head start (see Lucas and Crosier, 1982).

Let us define $Q_i=S_i^2/\sigma_0^2$ and $W=\hat{\sigma}_0^2/\sigma_0^2$. It is well known (see e.g. Chen, 1998) that Q_i and W are both gamma random variables of respective parameters $((n-1)/2,2\tau^2/(n-1))$ and $(m(n-1)/2,2/m(n-1))$. Therefore, the probability density

functions $f_Q(q)$ and $f_W(w)$ of Q_i and W are equal to

$$f_Q(q) = f_\gamma\left(q\middle|\frac{n-1}{2},\frac{2\tau^2}{n-1}\right),$$

$$f_W(w) = f_\gamma\left(w\middle|\frac{m(n-1)}{2},\frac{2}{m(n-1)}\right),$$

where

$$f_\gamma(x|a,b) = \frac{e^{-x/b}x^{a-1}}{b^a\Gamma(a)}.$$

It is also clear that $Q_i/W = S_i^2/\hat{\sigma}_0^2$ and Eq. (3) can thus be rewritten as

$$G_0^+ = g_0^+,$$

$$G_i^+ = \max(G_{i-1}^+ + \ln(Q_i/W) - k, 0), \quad i = 1,2,\ldots$$

and correspondingly Eq. (4) is equal to

$$G_0^- = g_0^-,$$

$$G_i^- = \min(G_{i-1}^- + \ln(Q_i/W) + k, 0), \quad i = 1,2,\ldots .$$

We are now ready to compute the exact run length distribution of the proposed CUSUM charts along with the average run length (ARL) and the standard deviation run length (SDRL). This is carried out in the following section.

3. The run length distribution of the CUSUM chart for the variance with estimated parameters

The control charts are generally evaluated in terms of their run length (or RL in short) distribution. The RL is defined as the number of samples until the statistic plotted on a control chart crosses one of the control limits. Let L_+ denote the random variable representing the RL of the upward CUSUM chart, i.e.

$$L_+ = \inf\{\ell \geq 1 | G_\ell^+ > h\}$$

and let

$$f_{L_+}(\ell|w,g_0^+) = P(L_+ = \ell | W = w, G_0^+ = g_0^+)$$

be the (conditional) probability mass function of L_+ given a specific value w for the random variable W and the starting value (constant) g_0^+.

Let L_- denote the random variable representing the RL of the downward CUSUM chart, i.e.

$$L_- = \inf\{\ell \geq 1 | G_\ell^- < -h\}$$

and let

$$f_{L_-}(\ell|w,g_0^-) = P(L_- = \ell | W = w, G_0^- = g_0^-)$$

be the (conditional) probability mass function of L_- given a specific value w for the random variable W and the starting value (constant) g_0^-.

For simplicity, the term $W=w$ will be removed in the rest of the paper when unnecessary. In this section, we will derive the main properties of the run length for the upward and downward chart, i.e. probability mass function, cumulative distribution function, mean (denoted *ARL*) and standard-deviation (denoted *SDRL*). We will first present an integral equation approach and then a Markov chain approach, both approaches being numerically solvable.

3.1. Integral equation approach

Jones et al. (2004) evaluated the performance of the CUSUM chart for the mean when the parameters are estimated using integral equations. We will use a similar approach in order to evaluate the performance of our upward and downward CUSUM charts for monitoring the variance.

The conditional probability mass function of L_+ can be computed using the following two propositions.

Proposition 1. *The conditional probability mass function of L_+ for $\ell = 1$ is equal to*

$$f_{L_+}(1|w,g_0^+) = 1 - F_Q(w\exp(h - g_0^+ + k)),$$

where F_Q is the cumulative distribution function of Q_i.

Proof. By definition $f_{L_+}(1|w,g_0^+) = P(L_+ = 1|G_0^+ = g_0^+)$. Since, the event $L_+ = 1$ is equivalent to the event $\ln(Q_1/w)+g_0^+ - k \geq h$, we may readily derive the following relation:

$$f_{L_+}(1|w,g_0^+) = P(\ln(Q_1/w)+g_0^+ - k \geq h).$$

Finally, after isolating Q_1, we have

$$f_{L_+}(1|w,g_0^+) = P(Q_1 \geq w\exp(h-g_0^+ + k)) = 1-F_Q(w\exp(h-g_0^+ + k)). \qquad \square$$

Proposition 2. *The conditional probability mass function of L_+ for $\ell = 2,3,\ldots$ is equal to*

$$f_{L_+}(\ell|w,g_0^+) = f_{L_+}(\ell-1|w,0)F_Q(w\exp(-g_0^+ + k)) + w\int_0^h f_{L_+}(\ell-1|w,c)\exp(c-g_0^+ + k)f_Q(w\exp(c-g_0^+ + k))\,\mathrm{d}c.$$

Proof. Given $G_0^+ = g_0^+$ the next value G_1^+ will be

- **either** $G_1^+ = 0 \Leftrightarrow \ln(Q_1/w)+g_0^+ - k \leq 0$,
- **or** $G_1^+ = \ln(q/w)+g_0^+ - k$ where $q \in \Xi$ is a particular value of Q_1 and where $\Xi = \{q \in \mathbb{R}|0 < \ln(q/w)+g_0^+ - k < h\}$.

Then, for $\ell = 2,3,\ldots$, we can split $f_{L_+}(\ell|w,g_0^+) = P(L_+ = \ell|G_0^+ = g_0^+)$ into two parts

$$f_{L_+}(\ell|w,g_0^+) = \overbrace{P(L_+ = \ell-1|G_0^+ = 0)}^{f_{L_+}(\ell-1|w,0)}P(\ln(Q_1/w)+g_0^+ - k \leq 0) + \int_\Xi \underbrace{P(L_+ = \ell-1|G_0^+ = \ln(q/w)+g_0^+ - k)}_{f_{L_+}(\ell-1|w,\ln(q/w)+g_0^+ - k)} f_Q(q)\,\mathrm{d}q.$$

Isolating Q_1 in $P(\ln(Q_1/w)+g_0^+ - k \leq 0)$ yields

$$P(\ln(Q_1/w)+g_0^+ - k \leq 0) = P(Q_1 \leq w\exp(-g_0^+ + k)) = F_Q(w\exp(-g_0^+ + k)).$$

If we introduce the variable $c = \ln(q/w)+g_0^+ - k$ then $q = w\exp(c-g_0^+ + k)$, $\mathrm{d}q = w\exp(c-g_0^+ + k)\,\mathrm{d}c$ and we finally deduce the following integral equation, for $\ell = 2,3,\ldots$:

$$f_{L_+}(\ell|w,g_0^+) = f_{L_+}(\ell-1|w,0)F_Q(w\exp(-g_0^+ + k)) + w\int_0^h f_{L_+}(\ell-1|w,c)\exp(c-g_0^+ + k)f_Q(w\exp(c-g_0^+ + k))\,\mathrm{d}c. \qquad \square$$

The corresponding propositions for computing the conditional probability mass function of L_- follow. Note that since the proofs use exactly the same steps as in Propositions 1 and 2 they are omitted.

Proposition 3. *The conditional probability mass function of L_- for $\ell = 1$ is equal to*

$$f_{L_-}(1|w,g_0^-) = F_Q(w\exp(-h-g_0^- - k)),$$

where F_Q is the cumulative distribution function of Q_i.

Proposition 4. *The conditional probability mass function of L_- for $\ell = 2,3,\ldots$ is equal to*

$$f_{L_-}(\ell|w,g_0^-) = f_{L_-}(\ell-1|w,0)(1-F_Q(w\exp(-g_0^- - k))) + w\int_{-h}^0 f_{L_-}(\ell-1|w,c)\exp(c-g_0^- - k)f_Q(w\exp(c-g_0^- - k))\,\mathrm{d}c.$$

Using Propositions 1 and 2 we are able to compute the conditional probability mass function of L_+. The unconditional probability mass function $f_{L_+}(\ell|g_0^+)$ of L_+ is then given by

$$f_{L_+}(\ell|g_0^+) = \int_0^\infty f_{L_+}(\ell|w,g_0^+)f_W(w)\,\mathrm{d}w$$

and the unconditional cumulative distribution function $F_{L_+}(\ell|g_0^+)$ of L_+ can immediately be derived from the previous formula

$$F_{L_+}(\ell|g_0^+) = \sum_{i=1}^{\ell} f_{L_+}(i|g_0^+).$$

The unconditional probability mass function and the unconditional cumulative distribution function of L_- are given by the same formulas just by replacing L_+ and g_0^+ with L_- and g_0^- respectively.

Let $ARL(w,g_0^+) = E(L_+|w,g_0^+)$ be the (conditional) average run length of the upward chart. We are now able to prove the following proposition.

Proposition 5. *The (conditional) average run length $ARL(w,g_0^+)$ of the upward chart can be obtained by solving the following integral equation*:

$$ARL(w,g_0^+) = 1 + ARL(w,0)F_Q(w\exp(-g_0^+ + k)) + w\int_0^h ARL(w,c)\exp(c-g_0^+ + k)f_Q(w\exp(c-g_0^+ + k))\,\mathrm{d}c.$$

Proof. By definition we have

$$ARL(w,g_0^+) = \sum_{\ell=1}^{\infty} \ell f_{L_+}(\ell|w,g_0^+).$$

Separating the case $\ell=1$, rearranging the remaining terms from $\ell=1$ and expanding $\ell+1$ gives

$$\begin{aligned} ARL(w,g_0^+) &= f_{L_+}(1|w,g_0^+) + \sum_{\ell=2}^{\infty} \ell f_{L_+}(\ell|w,g_0^+) = f_{L_+}(1|w,g_0^+) + \sum_{\ell=1}^{\infty} (\ell+1) f_{L_+}(\ell+1|w,g_0^+) \\ &= f_{L_+}(1|w,g_0^+) + \sum_{\ell=1}^{\infty} f_{L_+}(\ell+1|w,g_0^+) + \sum_{\ell=1}^{\infty} \ell f_{L_+}(\ell+1|w,g_0^+). \end{aligned}$$

The first two terms sum to 1 and we have

$$ARL(w,g_0^+) = 1 + \sum_{\ell=1}^{\infty} \ell f_{L_+}(\ell+1|w,g_0^+).$$

Since

$$f_{L_+}(\ell+1|w,g_0^+) = f_{L_+}(\ell|w,0)F_Q(w\exp(-g_0^+ + k)) + w\int_0^h f_{L_+}(\ell|w,c)\exp(c-g_0^+ + k) f_Q(w\exp(c-g_0^+ + k))\,dc$$

we straightforwardly deduce

$$ARL(w,g_0^+) = 1 + ARL(w,0)F_Q(w\exp(-g_0^+ + k)) + w\int_0^h ARL(w,c)\exp(c-g_0^+ + k) f_Q(w\exp(c-g_0^+ + k))\,dc. \qquad \square$$

Let us also define $E2RL(w,g_0^+) = E(L_+^2|w,g_0^+)$ be the (conditional) 2nd non-central moment of the run length L_+. Then, the following proposition holds.

Proposition 6. *The (conditional) 2nd non-central moment $E2RL(w,g_0^+)$ of the run length L_+ can be obtained by solving the following integral equation*:

$$\begin{aligned} E2RL(w,g_0^+) = 1 &+ 2\left[ARL(w,0)F_Q(w\exp(-g_0^+ + k)) + w\int_0^h ARL(w,c)\exp(c-g_0^+ + k) f_Q(w\exp(c-g_0^+ + k))\,dc\right] \\ &+ E2RL(w,0)F_Q(w\exp(-g_0^+ + k)) + w\int_0^h E2RL(w,c)\exp(c-g_0^+ + k) f_Q(w\exp(c-g_0^+ + k))\,dc. \end{aligned}$$

Proof. The demonstration for $E2RL(w,g_0^+) = E(L_+^2|w,g_0^+)$ is very similar to the demonstration of $ARL(w,g_0^+)$. By definition we have

$$E2RL(w,g_0^+) = \sum_{\ell=1}^{\infty} \ell^2 f_{L_+}(\ell|w,g_0^+).$$

Separating the case $\ell=1$, rearranging the remaining terms from $\ell=1$ and expanding $(\ell+1)^2$ gives

$$\begin{aligned} E2RL(w,g_0^+) &= f_{L_+}(1|w,g_0^+) + \sum_{\ell=2}^{\infty} \ell^2 f_{L_+}(\ell|w,g_0^+) = f_{L_+}(1|w,g_0^+) + \sum_{\ell=1}^{\infty} (\ell+1)^2 f_{L_+}(\ell+1|w,g_0^+) \\ &= f_{L_+}(1|w,g_0^+) + \sum_{\ell=1}^{\infty} f_{L_+}(\ell+1|w,g_0^+) + 2\sum_{\ell=1}^{\infty} \ell f_{L_+}(\ell+1|w,g_0^+) + \sum_{\ell=1}^{\infty} \ell^2 f_{L_+}(\ell+1|w,g_0^+). \end{aligned}$$

The first two terms sum again to 1 and we have

$$E2RL(w,g_0^+) = 1 + 2\sum_{\ell=1}^{\infty} \ell f_{L_+}(\ell+1|w,g_0^+) + \sum_{\ell=1}^{\infty} \ell^2 f_{L_+}(\ell+1|w,g_0^+).$$

Finally, if we expand $f_{L_+}(\ell+1|w,g_0^+)$, we obtain

$$\begin{aligned} E2RL(w,g_0^+) = 1 &+ 2\left[ARL(w,0)F_Q(w\exp(-g_0^+ + k)) + w\int_0^h ARL(w,c)\exp(c-g_0^+ + k) f_Q(w\exp(c-g_0^+ + k))\,dc\right] \\ &+ E2RL(w,0)F_Q(w\exp(-g_0^+ + k)) + w\int_0^h E2RL(w,c)\exp(c-g_0^+ + k) f_Q(w\exp(c-g_0^+ + k))\,dc. \end{aligned} \qquad \square$$

The (conditional) average run length and the (conditional) 2nd non-central moment of L_- are given in the following two propositions. The proofs of these propositions are omitted again since they can be easily obtained from the proofs of Propositions 5 and 6.

Proposition 7. *The (conditional) average run length $ARL(w,g_0^-)$ of the downward chart can be obtained by solving the following integral equation*:

$$ARL(w,g_0^-) = 1 + ARL(w,0)(1-F_Q(w\exp(-g_0^- - k))) + w\int_{-h}^{0} ARL(w,c)\exp(c-g_0^- - k) f_Q(w\exp(c-g_0^- - k))\,\mathrm{d}c.$$

Proposition 8. *The (conditional) 2nd non-central moment $E2RL(w,g_0^-)$ of the run length L_- can be obtained by solving the following integral equation*:

$$E2RL(w,g_0^-) = 1 + 2\left[ARL(w,0)(1-F_Q(w\exp(-g_0^- - k))) + w\int_{-h}^{0} ARL(w,c)\exp(c-g_0^- - k) f_Q(w\exp(c-g_0^- - k))\,\mathrm{d}c\right]$$
$$+ E2RL(w,0)(1-F_Q(w\exp(-g_0^- - k))) + w\int_{-h}^{0} E2RL(w,c)\exp(c-g_0^- - k) f_Q(w\exp(c-g_0^- - k))\,\mathrm{d}c.$$

The unconditional average run length $ARL(g_0^+)$ and 2nd non-central moment $E2RL(g_0^+)$ of L_+ are equal to

$$ARL(g_0^+) = \int_0^{\infty} ARL(w,g_0^+) f_W(w)\,\mathrm{d}w,$$

$$E2RL(g_0^+) = \int_0^{\infty} E2RL(w,g_0^+) f_W(w)\,\mathrm{d}w,$$

and the standard deviation run length $SDRL(g_0^+)$ of the upward chart is

$$SDRL(g_0^+) = \sqrt{E2RL(g_0^+) - ARL^2(g_0^+)}.$$

The unconditional average run length $ARL(g_0^-)$, the 2nd non-central moment $E2RL(g_0^-)$ of L_- and the standard deviation run length $SDRL(g_0^-)$ are given by the same formulas just by replacing g_0^+ with g_0^-.

The exact solutions of the integral equations for $ARL(w,g_0^+)$, $E2RL(w,g_0^+)$ and $ARL(w,g_0^-)$, $E2RL(w,g_0^-)$ can be approximated using numerical integration for specific value of W (see for instance Castagliola et al., 2006). Then the equations for $ARL(g_0^+)$, $E2RL(g_0^+)$ and $ARL(g_0^-)$, $E2RL(g_0^-)$ are solved numerically using a Gaussian quadrature for variable W.

3.2. Markov chain approach

Let us assume that we have a discrete-time Markov chain with $p+2$ states, where states $0,1,\ldots,p$ are transient and state $p+1$ is an absorbing one. The transition probability matrix $\mathbf{P}$ of this discrete-time Markov chain is

$$\mathbf{P} = \begin{pmatrix} \mathbf{Q} & \mathbf{r} \\ \mathbf{0}^T & 1 \end{pmatrix} = \begin{pmatrix} Q_{0,0} & Q_{0,1} & \cdots & Q_{0,p} & r_0 \\ Q_{1,0} & Q_{1,1} & \cdots & Q_{1,p} & r_1 \\ \vdots & \vdots & & \vdots & \vdots \\ Q_{p,0} & Q_{p,1} & \cdots & Q_{p,p} & r_p \\ 0 & 0 & \cdots & 0 & 1 \end{pmatrix},$$

where $\mathbf{Q}$ is the $(p+1,p+1)$ matrix of transient probabilities, $\mathbf{0}=(0,0,\ldots,0)^T$ and the $(p+1,1)$ vector $\mathbf{r}$ satisfies $\mathbf{r}=\mathbf{1}-\mathbf{Q1}$ (i.e. row probabilities must sum to 1) with $\mathbf{1}=(1,1,\ldots,1)^T$. Let $\mathbf{q}$ be the $(p+1,1)$ vector of initial probabilities associated with the $p+1$ transient states, i.e. $\mathbf{q}=(q_0,q_1,\ldots,q_p)^T$.

The number of steps L_+ (L_-) until the process reaches the absorbing state is a *Discrete PHase-type* (or DPH) random variable of parameters $(\mathbf{Q},\mathbf{q})$, see for instance Neuts (1981) or Latouche and Ramaswami (1999). In control chart terminology, the random variable L_+ (L_-) is also called the *run length*. The probability mass function $f_{DPH}(\ell|\mathbf{Q},\mathbf{q})$ and the cumulative distribution function $F_{DPH}(\ell|\mathbf{Q},\mathbf{q})$ of a DPH random variable L_+ (L_-) are defined for $\ell = 1,2,\ldots$ and are equal to

$$f_{DPH}(\ell|\mathbf{Q},\mathbf{q}) = \mathbf{q}^T\mathbf{Q}^{\ell-1}\mathbf{r},$$

$$F_{DPH}(\ell|\mathbf{Q},\mathbf{q}) = 1-\mathbf{q}^T\mathbf{Q}^{\ell}\mathbf{1}.$$

However, there is no simple formula for the central moment $\mu_i = E((L_+ - E(L_+))^i|\mathbf{Q},\mathbf{q})$ of order $i \geq 1$ of a DPH random variable L_+. On the other hand, there is a simple formula for the factorial moment $v_i = E(L_+(L_+-1)\cdots(L_+-i+1)|\mathbf{Q},\mathbf{q})$ of order $i \geq 1$, which is

$$v_i = i!\mathbf{q}^T(\mathbf{I}-\mathbf{Q})^{-i}\mathbf{Q}^{i-1}\mathbf{1}.$$

Specifically, when $i=1$, 2 we may easily conclude that

$$v_1 = \mathbf{q}^T(\mathbf{I}-\mathbf{Q})^{-1}\mathbf{1},$$

$$v_2 = 2\mathbf{q}^T(\mathbf{I}-\mathbf{Q})^{-2}\mathbf{Q}\mathbf{1}.$$

Now, one can readily establish a compact formula for the mean $E(L_+)$ and standard deviation $\sigma(L_+)$ of L_+

$$E(L_+) = v_1,$$

$$\sigma(L_+) = \sqrt{v_2 - v_1^2 + v_1}.$$

One will end up with the same formulas in the case of the downward chart if he replaces L_+ with L_-.

The properties (probability mass function $f_{L_+}(\ell|w,g_0^+)$ or $f_{L_-}(\ell|w,g_0^-)$, cumulative distribution function $F_{L_+}(\ell|w,g_0^+)$ or $F_{L_-}(\ell|w,g_0^-)$, $ARL(w,g_0^+)=E(L_+)$ or $ARL(w,g_0^-)=E(L_-)$ and $SDRL(w,g_0^+)=\sigma(L_+)$ or $SDRL(w,g_0^-)=\sigma(L_-)$, ...) of the run length L_+ or L_- of our chart can be numerically evaluated using the formulas presented above. This flexible and relatively easy to use procedure, originally proposed by Brook and Evans (1972), involves dividing the interval between 0 and h ($-h$ and 0 in the downward chart) into p subintervals of width 2δ, where $\delta = h/(2p)$. By definition, H_j, $j=1,\ldots,p$, represents the midpoint of the jth subinterval and $H_0=0$ corresponds to the "restart state" feature of our chart. When the number p of subintervals is sufficiently large, this finite approach provides an effective method that allows the run length properties to be accurately evaluated. In order to be able to do the computations we need to know the generic elements of $\mathbf{Q}$ and $\mathbf{q}$ that can be evaluated using the following propositions.

Proposition 9. *The generic element $Q_{i,j}$, $i=0,1,\ldots,p$, of the matrix $\mathbf{Q}$ of transient probabilities for the upward chart is equal to*

- *if $j=0$*

$$Q_{i,0} = F_Q(w\exp(-H_i+k));$$

Table 1
ARL_1 and $SDRL_1$ for $\tau = 0.5,0.6,0.7,0.8,0.9$ (downward case), $\tau = 1.2,1.4,1.6,1.8$ (upward case) using different combinations of $n=3,5,7,9$, $m = 10,20,40,80,\infty$ and (k^*,h^*) corresponding to the *known* parameter case.

τ	(k^*,h^*)	$m=10$	$m=20$	$m=40$	$m=80$	$m=\infty$
$n=3$						
0.5	(1.000, 8.515)	(12.6, 6.3)	(10.7, 4.6)	(10.2, 4.2)	(10.0, 4.1)	(9.8, 3.9)
0.6	(0.921, 9.440)	(26.1, 17.6)	(17.6, 9.2)	(16.0, 7.7)	(15.4, 7.2)	(14.9, 6.8)
0.7	(0.819, 11.134)	(85.7, 73.6)	(34.4, 22.2)	(27.3, 15.2)	(25.4, 13.4)	(24.0, 12.1)
0.8	(0.725, 13.614)	(653.2, 635.3)	(109.0, 90.4)	(58.2, 39.4)	(48.6, 29.7)	(43.1, 24.3)
0.9	(0.638, 17.720)	(21419.4, 21390.3)	(1344.3, 1313.0)	(247.6,215.0)	(135.0, 101.8)	(96.0, 62.3)
1.2	(0.000, 4.185)	(2092.1, 2087.0)	(97.8, 92.7)	(56.6, 51.4)	(47.2, 42.0)	(40.8, 35.5)
1.4	(0.276, 2.540)	(31.2, 28.6)	(19.6, 17.0)	(16.9, 14.3)	(15.9, 13.3)	(15.0, 12.5)
1.6	(0.666, 1.351)	(10.9, 9.7)	(9.1, 7.9)	(8.5, 7.3)	(8.2, 7.0)	(8.0, 6.7)
1.8	(0.744, 1.203)	(6.1, 5.2)	(5.6, 4.6)	(5.4, 4.4)	(5.3, 4.3)	(5.2, 4.2)
$n=5$						
0.5	(0.766, 3.525)	(5.3, 2.2)	(5.0, 2.0)	(4.9, 1.9)	(4.8, 1.9)	(4.8, 1.8)
0.6	(0.641, 4.195)	(9.0, 4.6)	(7.9, 3.6)	(7.6, 3.3)	(7.5, 3.2)	(7.3, 3.1)
0.7	(0.534, 5.098)	(21.0, 14.2)	(14.3, 7.7)	(13.0, 6.5)	(12.6, 6.1)	(12.2, 5.7)
0.8	(0.437, 6.482)	(131.1, 120.2)	(37.1, 26.0)	(27.2, 16.1)	(24.7, 13.8)	(23.1, 12.2)
0.9	(0.347, 9.037)	(9555.8, 9535.4)	(495.5, 474.0)	(113.0, 90.9)	(73.7, 51.4)	(58.1, 35.7)
1.2	(0.026, 3.522)	(511.7, 505.7)	(38.4, 32.4)	(27.5, 21.4)	(24.6, 18.6)	(22.5, 16.5)
1.4	(0.319, 1.662)	(11.5, 9.4)	(9.4, 7.3)	(8.7, 6.6)	(8.5, 6.3)	(8.2, 6.1)
1.6	(0.535, 1.052)	(5.2, 4.0)	(4.8, 3.6)	(4.6, 3.4)	(4.5, 3.4)	(4.5, 3.3)
1.8	(0.626, 0.883)	(3.2, 2.3)	(3.1, 2.2)	(3.1, 2.2)	(3.0, 2.1)	(3.0, 2.1)
$n=7$						
0.5	(0.698, 2.162)	(3.4, 1.4)	(3.3, 1.3)	(3.3, 1.3)	(3.3, 1.2)	(3.2, 1.2)
0.6	(0.562, 2.646)	(5.6, 2.6)	(5.3, 2.3)	(5.1, 2.2)	(5.0, 2.1)	(5.0, 2.1)
0.7	(0.450, 3.289)	(11.2, 6.5)	(9.3, 4.6)	(8.8, 4.1)	(8.5, 4.0)	(8.4, 3.8)
0.8	(0.350, 4.287)	(48.8, 40.7)	(21.7, 13.6)	(18.1, 10.1)	(17.0, 9.0)	(16.2, 8.3)
0.9	(0.258, 6.211)	(4201.2, 4184.9)	(221.5, 204.5)	(69.4, 52.1)	(51.4, 34.0)	(43.1, 25.7)
1.2	(0.090, 2.527)	(69.0, 64.1)	(23.0, 18.1)	(18.7, 13.8)	(17.3, 12.4)	(16.3, 11.3)
1.4	(0.354, 1.172)	(7.2, 5.5)	(6.4, 4.7)	(6.1, 4.4)	(6.0, 4.3)	(5.9, 4.1)
1.6	(0.500, 0.827)	(3.5, 2.5)	(3.4, 2.3)	(3.3, 2.3)	(3.3, 2.2)	(3.2, 2.2)
1.8	(0.589, 0.682)	(2.3, 1.5)	(2.3, 1.5)	(2.3, 1.4)	(2.3, 1.4)	(2.2, 1.4)
$n=9$						
0.5	(0.681, 1.505)	(2.6, 1.0)	(2.5, 1.0)	(2.5, 1.0)	(2.5, 1.0)	(2.5, 0.9)
0.6	(0.528, 1.910)	(4.2, 1.8)	(4.0, 1.7)	(3.9, 1.6)	(3.9, 1.6)	(3.8, 1.6)
0.7	(0.411, 2.419)	(7.8, 4.2)	(7.0, 3.3)	(6.7, 3.1)	(6.6, 3.0)	(6.4, 2.9)
0.8	(0.309, 3.214)	(26.3, 19.8)	(15.6, 9.1)	(13.7, 7.4)	(13.1, 6.8)	(12.7, 6.3)
0.9	(0.215, 4.788)	(1955.0, 1941.2)	(121.7, 107.4)	(50.0, 35.5)	(40.0, 25.6)	(34.8, 20.4)
1.2	(0.121, 1.984)	(30.9, 26.8)	(16.7, 12.5)	(14.4, 10.2)	(13.6, 9.4)	(12.9, 8.8)
1.4	(0.357, 0.939)	(5.4, 3.9)	(5.0, 3.5)	(4.8, 3.3)	(4.7, 3.2)	(4.6, 3.1)
1.6	(0.482, 0.681)	(2.8, 1.8)	(2.7, 1.7)	(2.6, 1.7)	(2.6, 1.7)	(2.6, 1.7)
1.8	(0.570, 0.550)	(1.9, 1.1)	(1.9, 1.1)	(1.9, 1.1)	(1.9, 1.1)	(1.8, 1.1)

- *if* $j=1,2,\ldots,p$

$$Q_{i,j} = F_Q(w\exp(H_j+\delta-H_i+k))-F_Q(w\exp(H_j-\delta-H_i+k))$$

and the generic element q_j *of vector* $\mathbf{q}$ *of initial probabilities is equal to*

$$q_j = \begin{cases} 1 & \text{if } H_j-\delta < g_0^+ < H_j+\delta, \\ 0 & \text{otherwise.} \end{cases}$$

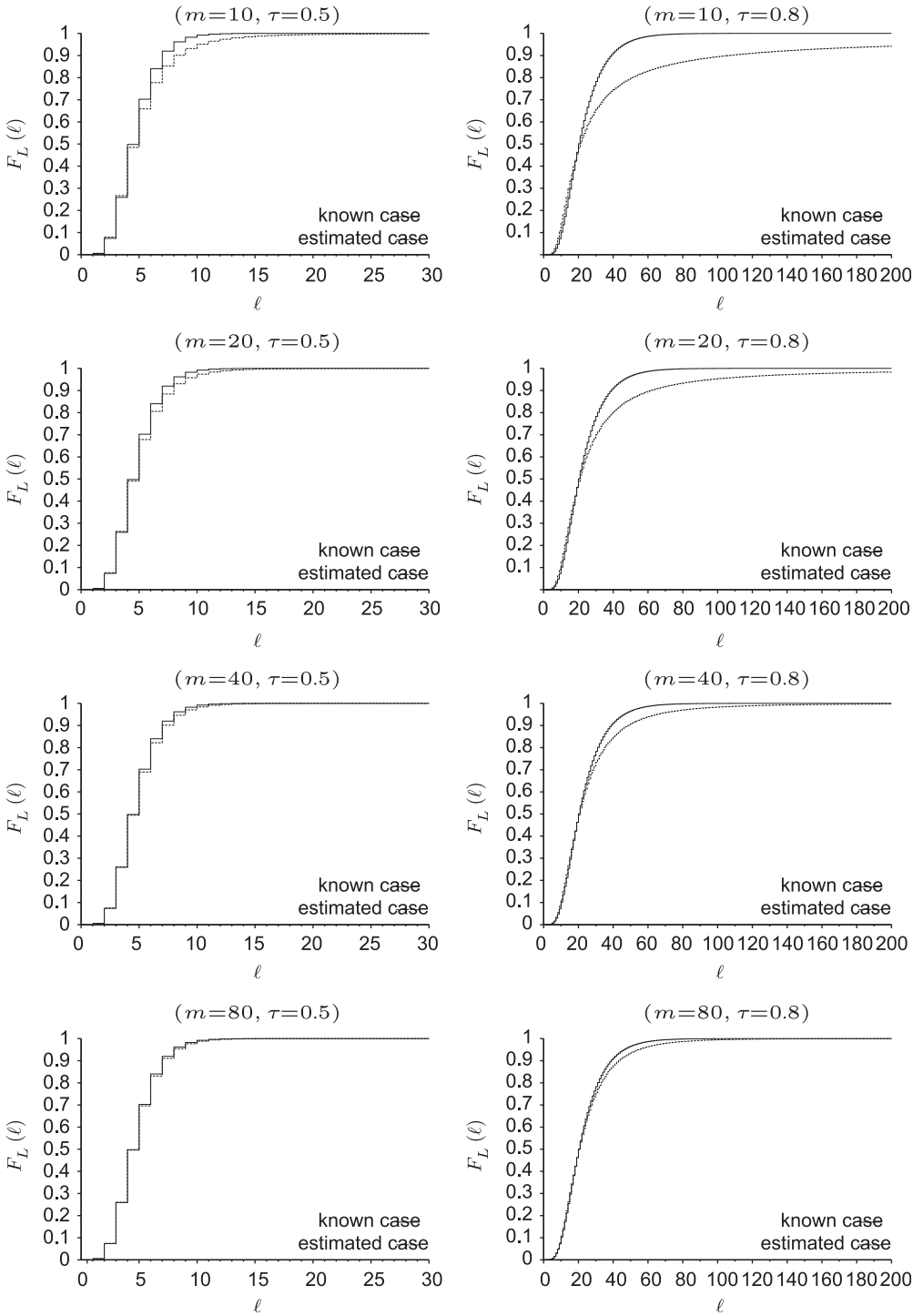


Fig. 1. c.d.f. of the RL for $n=5$, $\tau=0.5,0.8$, $m=10$, 20, 40, 80 (dotted line) and $m=\infty$ (plain line).

Proof. By definition, $Q_{i,j}$ is the probability that the Markov chain goes from state i to state j in one step. The probability $Q_{i,0}$ that the Markov chain goes from state i=0,1,…,p (i.e. starts from H_i) to state j=0 is

$$Q_{i,0} = P(\ln(Q/w) - k + H_i \leq 0).$$

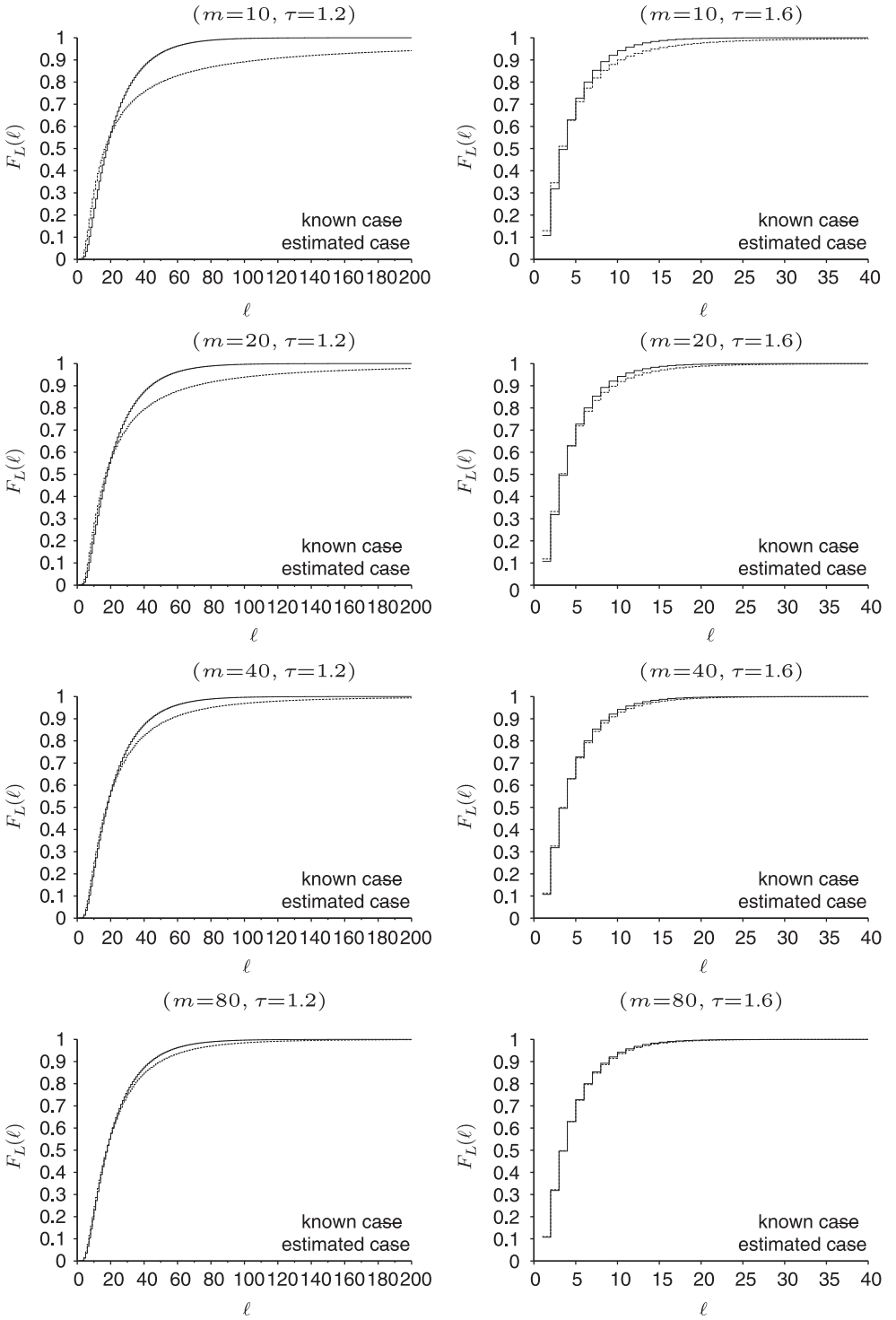


Fig. 2. c.d.f. of the RL for n=5, $\tau = 1.2,1.6$, m=10, 20, 40, 80 (dotted line) and $m = \infty$ (plain line).

Isolating Q gives

$$Q_{i,0} = P(Q \le w\exp(-H_i+k)) = F_Q(w\exp(-H_i+k)).$$

The probability $Q_{i,j}$ that the Markov chain goes from state i=0,1,…,p (i.e. starts from H_i) to state j=1,2…,p is

$$Q_{i,j} = P(H_j-\delta < \ln(Q/w)+H_i-k < H_j+\delta).$$

Splitting the probability into two parts yields

$$Q_{i,j} = P(\ln(Q/w)+H_i-k < H_j+\delta) - P(\ln(Q/w)+H_i-k < H_j-\delta).$$

Isolating Q again finally gives

$$\begin{aligned} Q_{i,j} &= P(Q \le w\exp(H_j+\delta-H_i+k)) - P(Q \le w\exp(H_j-\delta-H_i+k)) \\ &= F_Q(w\exp(H_j+\delta-H_i+k)) - F_Q(w\exp(H_j-\delta-H_i+k)). \end{aligned}$$ □

Proposition 10. *The generic element $Q_{i,j}$, i=0,1,…,p, of the matrix* **Q** *of transient probabilities for the downward chart is equal to*

- *if j=0*
$$Q_{i,0} = 1 - F_Q(w\exp(-H_i-k));$$
- *if j=1,2,…,p*
$$Q_{i,j} = F_Q(w\exp(H_j+\delta-H_i-k)) - F_Q(w\exp(H_j-\delta-H_i-k))$$

and the generic element q_j of vector **q** *of initial probabilities is equal to*

$$q_j = \begin{cases} 1 & \text{if } H_j-\delta < g_0^- < H_j+\delta, \\ 0 & \text{otherwise.} \end{cases}$$

The proof of Proposition 10 is not presented since it is a direct result from the proof of Proposition 9.

The unconditional probability mass function $f_{L_+}(\ell|g_0^+)$ or $f_{L_-}(\ell|g_0^-)$, cumulative distribution function $F_{L_+}(\ell|g_0^+)$ or $F_{L_-}(\ell|g_0^-)$, $ARL(g_0^+)$ or $ARL(g_0^-)$ and $SDRL(g_0^+)$ or $SDRL(g_0^-)$ can be obtained from the conditional ones by applying the same formulas presented in the previous subsection.

Table 2
ARL and *SDRL* for the downward CUSUM chart, for combination (k^*,h^*) satisfying ARL=370.4 when the process is in-control and the process parameters are assumed known.

(k^*,h^*)	m=10	m=20	m=40	m=80
n=3				
(0.600, 20.696)	(562031.0, 561993.0)	(32663.1, 32620.9)	(3620.5, 3575.2)	(1029.4, 981.9)
(0.700, 14.556)	(13351.2, 13331.1)	(2823.0, 2801.3)	(1042.2, 1019.4)	(613.9, 590.4)
(0.800, 11.534)	(3052.9, 3040.2)	(1184.4, 1171.0)	(674.9, 661.1)	(500.9, 486.8)
(0.900, 9.727)	(1481.5, 1472.7)	(786.3, 777.2)	(547.2, 537.9)	(451.5, 442.1)
n=5				
(0.400, 7.298)	(12194.9, 12181.0)	(2606.6, 2591.7)	(998.4, 982.9)	(603.0, 587.1)
(0.500, 5.494)	(2362.5, 2354.5)	(1015.1, 1006.8)	(622.8, 614.3)	(481.4, 472.8)
(0.600, 4.490)	(1162.2, 1157.0)	(684.2, 678.9)	(508.6, 503.2)	(435.0, 429.7)
(0.700, 3.842)	(803.0, 799.4)	(558.0, 554.3)	(457.4, 453.7)	(412.2, 408.6)
(0.800, 3.384)	(647.9, 645.3)	(496.1, 493.4)	(430.1, 427.5)	(399.5, 396.8)
(0.900, 3.039)	(567.3, 565.3)	(461.5, 459.4)	(414.2, 412.2)	(391.9, 389.9)
n=7				
(0.300, 5.129)	(13110.8, 13098.9)	(2593.6, 2581.0)	(982.8, 969.8)	(597.1, 583.8)
(0.400, 3.712)	(2153.8, 2147.4)	(949.2, 942.6)	(599.1, 592.5)	(471.7, 465.0)
(0.500, 2.961)	(1053.8, 1049.9)	(644.0, 640.0)	(492.0, 487.9)	(427.6, 423.5)
(0.600, 2.488)	(740.2, 737.6)	(531.9, 529.2)	(445.7, 443.0)	(406.7, 404.0)
(0.700, 2.157)	(608.7, 606.8)	(478.5, 476.6)	(421.9, 420.0)	(395.5, 393.6)
(0.800, 1.909)	(542.6, 541.1)	(449.7, 448.3)	(408.5, 407.1)	(389.1, 387.7)
(0.900, 1.711)	(505.9, 504.8)	(433.2, 432.1)	(400.7, 399.6)	(385.3, 384.2)
n=9				
(0.300, 3.316)	(3916.9, 3909.6)	(1290.8, 1283.2)	(694.7, 686.9)	(506.4, 498.5)
(0.400, 2.484)	(1293.5, 1289.4)	(714.7, 710.5)	(517.9, 513.7)	(438.5, 434.3)
(0.500, 2.009)	(803.3, 800.7)	(554.3, 551.7)	(455.0, 452.3)	(410.9, 408.3)
(0.600, 1.695)	(630.7, 629.0)	(486.9, 485.1)	(425.5, 423.7)	(397.2, 395.4)
(0.700, 1.467)	(552.6, 551.3)	(453.6, 452.3)	(410.2, 408.9)	(389.9, 388.6)
(0.800, 1.288)	(512.7, 511.7)	(435.8, 434.8)	(401.8, 400.8)	(385.8, 384.8)
(0.900, 1.139)	(490.9, 490.1)	(425.9, 425.1)	(397.1, 396.3)	(383.5, 382.7)

4. The performance of the one-sided CUSUM charts for the dispersion with estimated variance

The usual measures to evaluate the performance of control charts are the mean (*ARL*) and the standard deviation (*SDRL*) of the RL distribution. The average run length and the standard deviation run length when a process is in control are denoted ARL_0 and $SDRL_0$ respectively. On the other hand, the average run length and the standard deviation run length when a process is out-of-control are denoted ARL_1 and $SDRL_1$ respectively. A straightforward conclusion is that a control chart is considered better than its competitors if it has the smaller ARL_1 value for a specific shift τ when ARL_0 is the same for all the charts.

In Table 1, we present the ARL_1 and $SDRL_1$ values for a number of shifts $\tau = 0.5,0.6,0.7,0.8,0.9$ (for the downward case) and $\tau = 1.2,1.4,1.6,1.8$ (for the upward case) using different combinations of the sample size $n=3,5,7,9$ and number of samples $m = 10,20,40,80,\infty$. The case $m = \infty$ corresponds to the known parameter case and all the other m values to the estimated parameter case. The ARL_0 value is set to 370.4 for all the different cases considered and no head-start is used ($g_0^+ = g_0^- = 0$). For the computations we used the Markov chain methodology presented in Section 3.2. It must be noted that the two methodologies (integral equations or Markov chain) would give almost the same results with negligible differences.

The first column of Table 1 displays the shift in standard deviation. In the second column we have the optimal k and h values (k^* and h^* respectively) for the *known* parameter case (Chang and Gan's chart) for each shift. The optimality property here means that among all the different combinations of k and h values for the CUSUM chart under investigation, with the same ARL_0, the specific combination that is recorded in each case has the smallest ARL_1. As an example, when $n=3$ and $\tau = 1.4$, the optimal chart parameters (k^*,h^*) of the upward CUSUM chart for the variance with *known* process parameters are $k^*=0.276$ and $h^*=2.540$. In this case, the out-of-control *ARL* and *SDRL* are $ARL_1=15.0$ and $SDRL_1=12.5$. Now, if we use the same chart parameters $k^*=0.276$ and $h^*=2.540$ but assume that the process parameters μ_0 and σ_0 have been estimated based on $m=10$ Phase I samples, the out-of-control *ARL* and *SDRL* are $ARL_1=31.2$ and $SDRL_1=28.6$, i.e. the out-of-control *ARL* corresponding to the estimated parameter case is about twice the one corresponding to the known parameter case.

From the results we observe that for fixed n as m increases the *ARL* and *SDRL* values tend to the known parameter case ($m = \infty$). Furthermore, for fixed m an increasing value for n causes the *ARL* and *SDRL* to have lower values. We may easily observe that for small shifts (i.e. $0.8 \le \tau \le 1.2$) at least 80 samples are needed for the chart with estimated parameters to have a performance similar to the chart with known parameters. As the shift increases fewer samples are needed.

In Figs. 1 and 2, we may also observe the cumulative distribution functions for the estimated case (dotted line) together with the known parameter case (plain line) for several different combinations of m and τ when $n=5$. The results obtained here are consistent with the ones previously obtained in Chen (1998), Castagliola et al. (2009) and Maravelakis and Castagliola (2009), i.e. the run length distribution corresponding to the estimated parameter case is more skewed than the one

Table 3
ARL and *SDRL* for the upward CUSUM chart, for combination (k^*,h^*) satisfying $ARL=370.4$ when the process is in-control and the process parameters are assumed known.

(k^*,h^*)	$m=10$	$m=20$	$m=40$	$m=80$
$n=3$				
(0.200, 2.896)	(7702001.4, 7701997.7)	(6506.1, 6502.3)	(1061.6, 1057.7)	(588.8, 584.9)
(0.300, 2.439)	(2190697.0, 2190693.9)	(4673.4, 4670.3)	(973.4, 970.2)	(570.0, 566.8)
(0.400, 2.065)	(711213.9, 711211.4)	(3561.3, 3558.7)	(902.4, 899.8)	(553.6, 551.0)
(0.500, 1.753)	(265205.6, 265203.5)	(2869.6, 2867.5)	(848.1, 846.0)	(540.4, 538.2)
(0.600, 1.495)	(147020.0, 147018.2)	(2490.0, 2488.3)	(811.3, 809.5)	(530.8, 529.0)
(0.700, 1.283)	(93552.9, 93551.5)	(2203.0, 2201.6)	(778.4, 777.0)	(521.8, 520.3)
$n=5$				
(0.200, 2.195)	(3233458.6, 3271660.2)	(4534.3, 4530.3)	(995.4, 991.4)	(577.7, 573.6)
(0.300, 1.733)	(205274.0, 205271.1)	(2729.5, 2726.5)	(848.0, 845.0)	(541.8, 538.7)
(0.400, 1.388)	(45841.0, 45838.8)	(1960.0, 1957.7)	(756.9, 754.7)	(516.7, 514.4)
(0.500, 1.127)	(19086.7, 19085.1)	(1562.6, 1560.9)	(696.3, 694.5)	(498.4, 496.7)
(0.600, 0.928)	(8352.5, 8351.3)	(1266.8, 1265.6)	(643.9, 642.6)	(481.8, 480.5)
(0.700, 0.770)	(4766.8, 4765.9)	(1061.0, 1060.1)	(600.2, 599.3)	(466.9, 466.0)
$n=7$				
(0.200, 1.780)	(365723.1, 368305.4)	(3373.1, 3369.2)	(920.0, 916.1)	(561.2, 557.2)
(0.300, 1.348)	(36643.6, 36641.0)	(1957.5, 1954.8)	(761.7, 759.0)	(518.7, 516.0)
(0.400, 1.045)	(11948.6, 11946.7)	(1418.2, 1416.3)	(674.0, 672.1)	(491.8, 489.9)
(0.500, 0.828)	(5045.3, 5044.0)	(1100.8, 1099.5)	(610.5, 609.1)	(470.7, 469.3)
(0.600, 0.666)	(2931.4, 2930.5)	(906.7, 905.8)	(563.0, 562.1)	(453.6, 452.7)
(0.700, 0.534)	(2192.9, 2192.2)	(803.9, 803.2)	(533.3, 532.6)	(442.1, 441.4)
$n=9$				
(0.200, 1.502)	(103732.8, 103860.4)	(2673.9, 2670.3)	(856.5, 852.8)	(545.6, 541.9)
(0.300, 1.103)	(15360.5, 15358.1)	(1564.5, 1562.1)	(702.6, 700.2)	(501.2, 498.8)
(0.400, 0.836)	(5365.3, 5363.7)	(1131.0, 1129.4)	(617.9, 616.3)	(473.4, 471.7)
(0.500, 0.652)	(2687.2, 2686.1)	(886.4, 885.3)	(558.3, 557.3)	(452.0, 450.9)
(0.600, 0.511)	(1881.5, 1880.7)	(763.6, 762.8)	(522.0, 521.2)	(437.8, 437.1)
(0.700, 0.393)	(1589.3, 1588.7)	(706.7, 706.0)	(503.1, 502.5)	(430.0, 429.4)

corresponding to the known parameter case. The run length distribution becomes highly skewed when m is small and when τ is close to one. These figures also confirm our previous conclusion that as the shift or the number of samples increases the results for the estimated case tend to the known parameter case but, when the shift is small (i.e. $0.8 \leq \tau \leq 1.2$), the convergence requires a large number of samples.

It can be noted that, in Table 1, for some value of τ close to one, the out-of-control *ARL* and *SDRL* can be very large. For example, if $n=3$, $\tau = 0.9$ and $m=10$, the out-of-control *ARL* and *SDRL* are $ARL_1=21419.4$ and $SDRL_1=21390.3$. This is due to the fact that the run length distribution of the CUSUM chart with estimated parameters is highly skewed when m is small and when τ is close to one. This also corroborates the fact that using the chart constants corresponding to the known parameter case in the estimated parameter case can yield to in-control *ARL* and *SDRL* much larger than expected, making the constants corresponding to the known parameter case useless whatever the number of samples m or the sample size n. In Tables 2 and 3, all the combinations of (k^*,h^*) listed in the first column, for $n=3,5,7,9$, are such that $ARL=370.4$ when the process is in-control, i.e. $\tau = 1$, and when the parameters are assumed known. Table 2 records the values (k^*,h^*) corresponding to the downward CUSUM chart while Table 3 records the values (k^*,h^*) corresponding to the upward CUSUM chart. For instance, concerning the upward CUSUM chart, for $n=5$, the combination $(k^*=0.4, h^*=1.388)$ gives $ARL=370.4$ when the process is in-control and when the parameters are assumed known ($m=\infty$) but, when the parameters are estimated, we have $ARL=45841.0$ for $m=10$, $ARL=1960.0$ for $m=20$, $ARL=756.9$ for $m=40$ and $ARL=516.7$ for $m=80$. As we can see, when the process is in-control, there is a huge difference in this case between the known parameter case and the estimated parameter case. When k^* is small, the value of *ARL* in the estimated parameter case is large (and sometimes really large). In this case, the corresponding run length distribution is strongly skewed. When k^* increases, the value of *ARL* decreases but still remains far from the value $ARL=370.4$ corresponding to the known parameter case. The same conclusions hold for the *SDRL*.

In Tables 4 and 5, we present the optimal k and h parameters for different combinations of sample size $n=3, 5, 7, 9$, number of samples $m=10, 20, 40, 80$ and shift $\tau = 0.5,0.6,0.7,0.8,0.9$ (for the downward case) and $\tau = 1.2,1.4,1.6,1.8$ (for the upward case). For each combination of the n, m and τ, we record in the first line the optimal k^* and h^* values and in the second line the

Table 4
Optimal combination (k^*,h^*) and ARL_1 and $SDRL_1$ when $n=3$, 5 and $m=10$, 20, 40, 80.

τ	$m=10$	$m=20$	$m=40$	$m=80$
$n=3$				
0.5	(0.494, 10.342)	(0.817, 8.955)	(0.935, 8.515)	(0.987, 8.332)
	(8.7, 2.8)	(9.3, 3.5)	(9.6, 3.8)	(9.7, 4.0)
0.6	(0.031, 15.472)	(0.633, 11.020)	(0.793, 10.094)	(0.860, 9.741)
	(11.7, 3.2)	(13.6, 5.3)	(14.3, 6.1)	(14.6, 6.5)
0.7	(–, –)	(0.404, 14.983)	(0.652, 12.484)	(0.742, 11.731)
	(–, –)	(20.1, 7.4)	(22.3, 10.0)	(23.2, 11.1)
0.8	(–, –)	(–, –)	(0.487, 17.227)	(0.624, 14.982)
	(–, –)	(–, –)	(37.3, 16.7)	(40.5, 20.9)
0.9	(–, –)	(–, –)	(–, –)	(0.479, 22.798)
	(–, –)	(–, –)	(–, –)	(85.1, 44.6)
1.2	(0.000, 2.136)	(0.000, 2.854)	(0.000, 3.408)	(0.000, 3.761)
	(21.9, 19.8)	(28.8, 25.8)	(34.0, 30.1)	(37.2, 32.7)
1.4	(0.000, 2.136)	(0.000, 2.854)	(0.117, 2.790)	(0.196, 2.667)
	(8.3, 6.5)	(10.8, 8.3)	(12.8, 10.1)	(13.9, 11.2)
1.6	(0.287, 1.412)	(0.463, 1.399)	(0.562, 1.379)	(0.613, 1.366)
	(5.0, 3.9)	(6.2, 5.1)	(7.0, 5.8)	(7.5, 6.3)
1.8	(0.360, 1.277)	(0.542, 1.250)	(0.640, 1.230)	(0.691, 1.217)
	(3.6, 2.6)	(4.2, 3.3)	(4.7, 3.7)	(4.9, 3.9)
$n=5$				
0.5	(0.514, 3.949)	(0.650, 3.711)	(0.710, 3.613)	(0.739, 3.567)
	(4.5, 1.5)	(4.6, 1.7)	(4.7, 1.8)	(4.7, 1.8)
0.6	(0.293, 5.127)	(0.489, 4.575)	(0.569, 4.370)	(0.606, 4.278)
	(6.4, 2.2)	(6.9, 2.7)	(7.1, 2.9)	(7.2, 3.0)
0.7	(–, –)	(0.326, 5.961)	(0.440, 5.469)	(0.488, 5.273)
	(–, –)	(10.8, 4.3)	(11.5, 5.0)	(11.9, 5.4)
0.8	(–, –)	(0.099, 9.320)	(0.304, 7.464)	(0.376, 6.911)
	(–, –)	(17.8, 6.2)	(20.6, 9.4)	(21.9, 10.8)
0.9	(–, –)	(–, –)	(0.048, 16.183)	(0.249, 10.802)
	(–, –)	(–, –)	(42.3, 13.8)	(51.5, 26.9)
1.2	(0.000, 1.820)	(0.000, 2.451)	(0.000, 2.978)	(0.000, 3.342)
	(12.9, 10.6)	(16.1, 12.6)	(18.6, 14.0)	(20.4, 14.9)
1.4	(0.000, 1.820)	(0.116, 1.909)	(0.214, 1.799)	(0.266, 1.734)
	(5.2, 3.3)	(6.5, 4.3)	(7.3, 5.1)	(7.7, 5.6)
1.6	(0.261, 1.130)	(0.394, 1.097)	(0.464, 1.076)	(0.500, 1.063)
	(3.3, 2.2)	(3.8, 2.7)	(4.1, 3.0)	(4.3, 3.1)
1.8	(0.351, 0.971)	(0.487, 0.929)	(0.557, 0.905)	(0.592, 0.894)
	(2.4, 1.5)	(2.7, 1.8)	(2.8, 1.9)	(2.9, 2.0)

Table 5
Optimal combination (k^*,h^*) and ARL_1 and $SDRL_1$ when n=7, 9 and m=10, 20, 40, 80.

τ	m=10	m=20	m=40	m=80
n=7				
0.5	(0.532, 2.334)	(0.619, 2.241)	(0.660, 2.201)	(0.679, 2.181)
	(3.1, 1.1)	(3.2, 1.2)	(3.2, 1.2)	(3.2, 1.2)
0.6	(0.334, 3.028)	(0.457, 2.815)	(0.512, 2.726)	(0.537, 2.684)
	(4.5, 1.6)	(4.8, 1.8)	(4.9, 2.0)	(4.9, 2.0)
0.7	(0.116, 4.249)	(0.307, 3.673)	(0.383, 3.463)	(0.417, 3.373)
	(6.8, 2.3)	(7.6, 3.1)	(8.0, 3.5)	(8.2, 3.6)
0.8	(–, –)	(0.132, 5.449)	(0.256, 4.753)	(0.305, 4.501)
	(–, –)	(13.2, 5.1)	(14.8, 6.8)	(15.5, 7.5)
0.9	(–, –)	(–, –)	(0.077, 8.925)	(0.186, 7.131)
	(–, –)	(–, –)	(33.3, 13.4)	(38.7, 20.2)
1.2	(0.000, 1.628)	(0.000, 2.200)	(0.000, 2.700)	(0.038, 2.685)
	(9.5, 7.1)	(11.7, 8.0)	(13.6, 8.8)	(14.9, 9.9)
1.4	(0.089, 1.342)	(0.221, 1.264)	(0.287, 1.220)	(0.320, 1.196)
	(4.1, 2.5)	(4.9, 3.2)	(5.3, 3.6)	(5.6, 3.9)
1.6	(0.279, 0.905)	(0.389, 0.868)	(0.444, 0.849)	(0.472, 0.839)
	(2.6, 1.6)	(2.9, 1.8)	(3.1, 2.0)	(3.1, 2.1)
1.8	(0.373, 0.757)	(0.481, 0.719)	(0.535, 0.700)	(0.562, 0.691)
	(1.9, 1.1)	(2.1, 1.2)	(2.1, 1.3)	(2.2, 1.4)
n=9				
0.5	(0.562, 1.584)	(0.623, 1.544)	(0.653, 1.524)	(0.667, 1.514)
	(2.4, 0.9)	(2.4, 0.9)	(2.5, 0.9)	(2.5, 0.9)
0.6	(0.357, 2.115)	(0.447, 2.005)	(0.489, 1.955)	(0.509, 1.931)
	(3.5, 1.3)	(3.7, 1.4)	(3.8, 1.5)	(3.8, 1.5)
0.7	(0.166, 2.933)	(0.301, 2.639)	(0.359, 2.522)	(0.385, 2.469)
	(5.5, 2.0)	(6.0, 2.4)	(6.2, 2.7)	(6.3, 2.8)
0.8	(–, –)	(0.145, 3.863)	(0.235, 3.492)	(0.274, 3.344)
	(–, –)	(10.7, 4.3)	(11.7, 5.3)	(12.2, 5.8)
0.9	(–, –)	(–, –)	(0.080, 6.299)	(0.158, 5.362)
	(–, –)	(–, –)	(27.9, 12.1)	(31.5, 16.5)
1.2	(0.000, 1.493)	(0.000, 2.021)	(0.037, 2.181)	(0.080, 2.080)
	(7.8, 5.3)	(9.5, 5.9)	(11.1, 6.8)	(12.0, 7.8)
1.4	(0.157, 1.020)	(0.255, 0.987)	(0.306, 0.964)	(0.331, 0.952)
	(3.5, 2.1)	(4.0, 2.5)	(4.3, 2.8)	(4.5, 3.0)
1.6	(0.296, 0.748)	(0.389, 0.715)	(0.436, 0.698)	(0.459, 0.689)
	(2.2, 1.3)	(2.4, 1.4)	(2.5, 1.5)	(2.5, 1.6)
1.8	(0.392, 0.612)	(0.481, 0.581)	(0.526, 0.565)	(0.548, 0.557)
	(1.6, 0.9)	(1.7, 1.0)	(1.8, 1.0)	(1.8, 1.0)

ARL_1 and $SDRL_1$ values. For instance, when n=7, m=20 and $\tau = 1.8$ the optimal parameters are k^*=0.481, h^*=0.719 and ARL_1=2.1, $SDRL_1$=1.2. In Tables 4 and 5, as in Table 1, we have set ARL_0=370.4 and we used again the Markov chain methodology of Section 3.2. It must be noted that for some combinations of n, m and τ it was not possible to compute the optimal parameters and their *ARL* and *SDRL* values. This is due to the "divergence" of the Markov chain algorithm that computes *ARL*'s corresponding to too small values for h and too large values for k (this problem is also encountered with EWMA type control charts when the smoothing parameter λ is too small).

The first conclusion drawn from Tables 4 and 5 is that for fixed n the reference value k increases as m increases. The control limit h behaves in the same manner for fixed n but only when $\tau \geq 1.6$. For fixed m as the shift increases so does the reference value k. It must be noted that the ARL_1 and $SDRL_1$ values in Tables 4 and 5 are not directly comparable since they are computed for different m and n values for different combination of the parameters k and h.

In Table 6 we present a comparison between three control charts for the variance, for n=5,9, in the case where the parameters are estimated. For each shift $\tau = 1.2,1.4,1.6,1.8$ and for each number of sample m=10,20,40,80, values of (ARL_1,$SDRL_1$) are presented for

- The S^2 chart with estimated parameters investigated in Castagliola et al. (2009).
- The EWMA S^2 chart with estimated parameters investigated in Maravelakis and Castagliola (2009).
- The CUSUM S^2 chart with estimated parameters proposed in this paper.

For example, if n=5, $\tau = 1.4$ and m=10, the values for (ARL_1,$SDRL_1$) are

- (28.1, 55.9) for the S^2 chart with estimated parameters.
- (5.2, 3.3) for the EWMA S^2 chart with estimated parameters.
- (5.2, 3.3) for the CUSUM S^2 chart with estimated parameters.

Table 6
Comparison of (ARL_1, $SDRL_1$) for three control charts for the variance with estimated parameters (S^2 chart, EWMA S^2 chart, CUSUM S^2 chart).

τ	$m=10$	$m=20$	$m=40$	$m=80$	
$n=5$					
1.2	(117.4, 212.0)	(92.3, 143.0)	(78.3, 102.4)	(71.3, 82.2)	S^2
	(13.1, 10.7)	(16.3, 12.8)	(18.9, 14.4)	(20.5, 15.3)	EWMA S^2
	(12.9, 10.6)	(16.1, 12.6)	(18.6, 14.0)	(20.4, 14.9)	CUSUM S^2
1.4	(28.1, 55.9)	(21.7, 30.9)	(19.1, 22.4)	(17.9, 19.1)	S^2
	(5.2, 3.3)	(6.5, 4.2)	(7.4, 5.0)	(7.8, 5.4)	EWMA S^2
	(5.2, 3.3)	(6.5, 4.3)	(7.3, 5.1)	(7.7, 5.6)	CUSUM S^2
1.6	(9.8, 15.4)	(8.4, 9.9)	(7.8, 8.1)	(7.5, 7.4)	S^2
	(3.3, 2.2)	(3.9, 2.7)	(4.3, 3.0)	(4.4, 3.2)	EWMA S^2
	(3.3, 2.2)	(3.8, 2.7)	(4.1, 3.0)	(4.3, 3.1)	CUSUM S^2
1.8	(5.0, 6.2)	(4.6, 4.7)	(4.4, 4.1)	(4.3, 3.9)	S^2
	(2.4, 1.5)	(2.7, 1.8)	(2.9, 2.0)	(3.0, 2.1)	EWMA S^2
	(2.4, 1.5)	(2.7, 1.8)	(2.8, 1.9)	(2.9, 2.0)	CUSUM S^2
$n=9$					
1.2	(74.4, 138.5)	(56.6, 84.6)	(48.2, 59.7)	(44.3, 49.1)	S^2
	(7.8, 5.4)	(9.6, 6.0)	(11.1, 6.8)	(12.1, 7.7)	EWMA S^2
	(7.8, 5.3)	(9.5, 5.9)	(11.1, 6.8)	(12.0, 7.8)	CUSUM S^2
1.4	(12.6, 20.3)	(10.5, 12.7)	(9.6, 10.2)	(9.2, 9.2)	S^2
	(3.5, 2.0)	(4.1, 2.5)	(4.4, 2.8)	(4.6, 2.9)	EWMA S^2
	(3.5, 2.1)	(4.0, 2.5)	(4.3, 2.8)	(4.5, 3.0)	CUSUM S^2
1.6	(4.5, 5.2)	(4.1, 4.0)	(3.9, 3.5)	(3.8, 3.3)	S^2
	(2.2, 1.3)	(2.5, 1.5)	(2.6, 1.6)	(2.7, 1.7)	EWMA S^2
	(2.2, 1.3)	(2.4, 1.4)	(2.5, 1.5)	(2.5, 1.6)	CUSUM S^2
1.8	(2.5, 2.2)	(2.3, 1.9)	(2.3, 1.8)	(2.3, 1.7)	S^2
	(1.7, 0.9)	(1.8, 1.0)	(1.8, 1.1)	(1.9, 1.1)	EWMA S^2
	(1.6, 0.9)	(1.7, 1.0)	(1.8, 1.0)	(1.8, 1.0)	CUSUM S^2

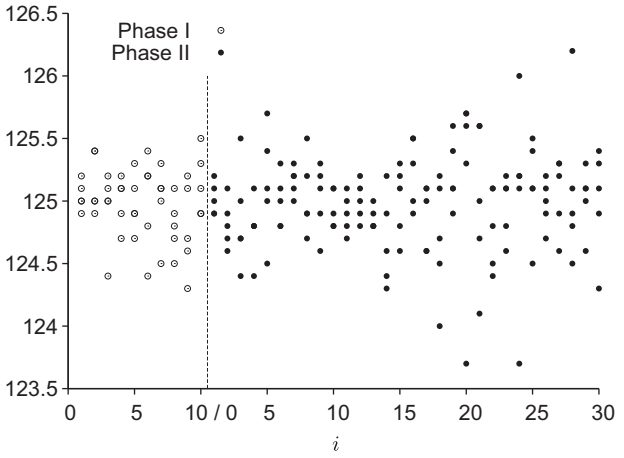


Fig. 3. Phase I and II data sets.

The results concerning $\tau < 1$ are not available for the EWMA S^2 chart with estimated parameters investigated in Maravelakis and Castagliola (2009). For this reason, they are not presented here. As it can be noticed in Table 6, as in the known parameter case, both the EWMA S^2 and CUSUM S^2 charts outperform the S^2 chart and the difference between the EWMA S^2 chart and the CUSUM S^2 chart is almost negligible, with a tiny superiority for the CUSUM S^2. These results should also hold for $\tau < 1$.

5. An example

In order to illustrate the use of the CUSUM control chart, let us consider a procedure to be designed with the aim of monitoring the variability within a yogurt cup filling process: a critical to quality parameter of the filling process is the deviation of each cup filling from a specified target $\mu_0 = 125$ g. We want to monitor the process dispersion by means of the upward CUSUM control chart introduced in Eq. (3). The Phase I data set used in this example consists of $m=10$ subgroups of

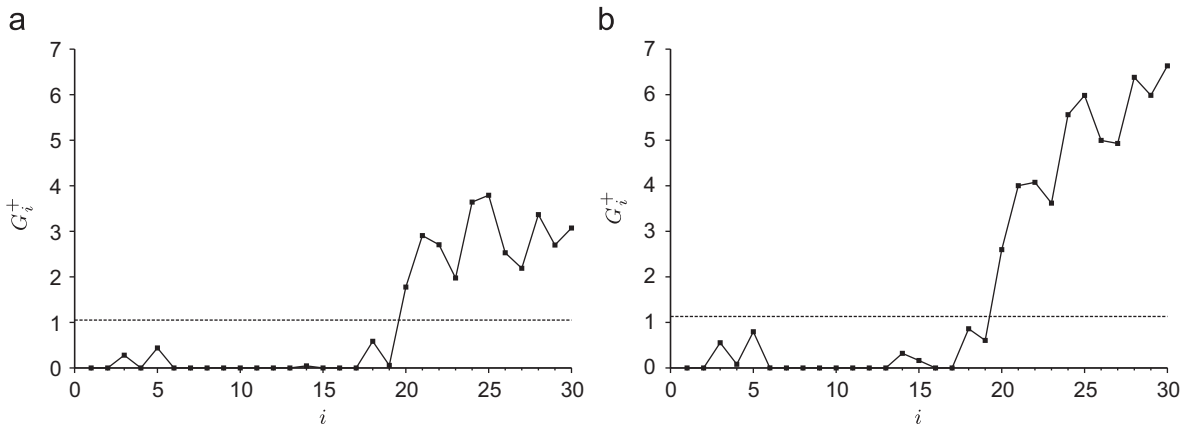


Fig. 4. Upward CUSUM chart for S^2 in (a) known parameter case, (b) estimated parameter case.

size $n=5$ plotted in the left part of Fig. 3 with "$\circ$" . From this Phase I data set we deduce that an estimate of the mean is $\hat{\mu}_0 = 124.982$ and $\hat{\sigma}_0^2 = 0.0766$. Concerning the upward CUSUM chart, we have chosen to select:

- the optimal parameters $k^*=0.535$ and $h^*=1.052$ optimally designed to detect a shift $\tau = 1.6$ when σ_0 is assumed known (see Table 1).
- the optimal parameters $k^*=0.261$ and $h^*=1.13$ optimally designed to detect a shift $\tau = 1.6$ when σ_0 is estimated from a Phase I data set consisting of $m=10$ subgroups of size $n=5$ (see Table 4).

The Phase II data set used in this example consists of $m=30$ subgroups of size $n=5$ plotted in the right part of Fig. 3 with "$\bullet$" . The first 15 subgroups are supposed in-control while the last 15 subgroups are supposed to have a larger variability and to be out-of-control. In both Fig. 4(a) and (b) we plotted the upward CUSUM chart corresponding to the Phase II data set. In Fig. 4(a) we used the optimal parameters $k^*=0.535$ and $h^*=1.052$ defined above in the σ_0^2 known case while, in Fig. 4(b), we used the optimal parameters $k^*=0.261$ and $h^*=1.13$ defined above in the σ_0^2 estimated case. In both cases, these control charts signal several out-of-control situations (the first one for the 20th subgroup) and the two proposed control charts behave similarly. As we can see, the control limits ($h^*=1.052$ and 1.13) are almost the same but, the values of k^* are really different ($k^*=0.535$ and 0.261). The fact that the value of k^* is smaller in the estimated parameter case than in the known parameter case allows the statistic G_i^+ to shift away more rapidly in the estimated parameter case than in the known parameter case.

6. Conclusions

Recent research has shown that the performance of control charts is seriously affected if the process parameters are estimated. In this paper we have presented results on the performance of the CUSUM control chart for monitoring the process dispersion when the variance is estimated using an integral equation and a Markov chain approach. It is shown that more samples are needed for the chart with estimated variance to behave like the one with known variance. Additionally, numerical results are presented for choosing the optimal parameters for each combination of sample size, number of samples and shift.

All the results in this paper are based on the assumptions of normality and independence. In case these assumptions do not hold the practitioner should not take into account the presented results.

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